SOWMATHI R

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CAREER OBJECTIVE

Seeking a position in a progressive organization where I can leverage my expertise in Electronics and Communication Engineering, particularly in Embedded Systems, IoT, PCB Design, VLSI, Fullstack Development and Software Coding Skills, while continuously advancing my skills through practical experience and contributing to cutting-edge technological solutions.

EDUCATION

Akshaya College of Engineering and Technology	B.E ECE	CGPA: 6.96 (Upto 5th Semester)	2023-2027
Government Girls Higher Secondary School	HSC	78%	2021-2022
Government Girls Higher Secondary School	SSLC	100%	2020-2021

TECHNICAL SKILLS

Programming Language: Verilog, System Verilog, Python, C++, HTML, CSS and Embedded C.

Hardware and Software Skills: VLSI, Embedded Systems and IoT, PCB Design and Fullstack Development.

Tools: Xilinx, Eda Playground, Arduino, MATLAB and KiCad.

Soft Skills: Problem-Solving, Communication, Leadership.

INTERNSHIP

Company Name: Gateway Software Solutions July (2026)

The internship focuses on understanding how hardware and software integrate to create intelligent, connected devices. The program combined hands-on hardware prototyping with IoT-based application development, enabling practical exposure to real-world problem-solving.

Company Name: Taras Systems and Solution. Dec-May (2026)

The internship focuses on the transition from basic hardware design to advanced verification methodologies like Verilog, System Verilog and UVM technology.

Company Name: Enthu Tech Solution India Pvt. Ltd. Dec (2025)

The internship focuses on designing and fabricating PCBs for electronic projects. Utilized expertise in PCB design software and fabrication technique.

Company Name: Nxt Logic Software Solutions Feb-Mar (2024)

The internship builds practical skills in front-end and back-end programming by converting design concepts into functional websites and applications.

PROJECTS

Title: Personal Portfolio Website (2026)

Designed and developed a responsive personal portfolio website using HTML and CSS in Visual Studio Code to showcase my profile, skills, projects, education, and achievements.

Title: Smart Multi-Product Vending Machine with Automatic Change Return Using System Verilog HDL (2026)

The project implements a Smart Multi-Product Vending Machine using Verilog HDL and UVM. It performs product selection, money checking, automatic dispensing, and change return using FSM-based digital design and functional verification techniques.

Title: IoT Enabled Smart Lock for Home Security (2026)

Created a secure, automated door access system using RFID technology for digital authentication and improved home security.

Title: AI-Enabled Prosthetic Hand with Tremor Monitoring and Quantum Cryptography Security (2026)

Real-time edge-AI based wearable system that not only detects Parkinson's tremors, but also predicts changes in tremor severity and delivers adaptive vibration feedback through a closed-loop mechanism, enabling proactive, personalized, and efficient tremor management.

Title: Traffic Management using AI (2024)

AI-powered traffic management systems utilize real-time data analytics, machine learning, and IoT sensors to optimize traffic flow, reduce congestion, and improve safety. These systems can predict traffic patterns, detect incidents, and dynamically adjust traffic signal timings to minimize congestion and reduce travel times.

Title: Mobile Control Robot (2023)

A Mobile controlled robot is a wireless robot that can be operated using a smartphone via Bluetooth or Wi-Fi. It allows remote navigation and control, often used in automation, surveillance, or educational project

CERTIFICATIONS

- Certified in VLSI Design and Verification. (2026)
- Certified in VLSI Frontend End Design. (2025)
- Certified in AI Foundation Associate - Oracle University. (2025)
- Certified in MangoDB Basics. (2025)
- Certified in Embedded Systems with IoT. (2025)
- Certified in PCB Design and Fabrications. (2025)
- Certified in Artificial Intelligence and Internet of Things. (2025)
- Certified in Web Development. (2024)

APPRECIATIONS

- Participated in Smart India Hackathon (2024,2025)
- Participated in Accenture Hackathon (2024)
- Participated in Tech Day (2025,2026)